ECOM90009 Quantitative Methods for Business

Workshop 3 (Week 4)

Zheng Fan

The University of Melbourne

Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Assignment

Assignment 1

- worth up to 10% of your final grade for this subject.
- Register your group by **Thursday, March 26, at 11:59 pm**.
- Submit by **Thursday, April 2, at 11.59 pm**.
- The word limit for this assignment is 1,000 words.
- Note that the "AI Usage Declaration" is not included in the word count.

Introduction

- Probability and discrete distributions
- Bivariate distribution
- Marginal, conditional and joint probabilities
- Independent and mutually exclusive events
- Covariance of two discrete random variables

Question 1

Indicate whether each statement below is true or false. Justify your answer.

a. Assume that A and B are independent events, with $P(A) = 0.30$ and $P(B) = 0.50$. The probability that both events will occur simultaneously is 0.80.

b. The probability of the union of two mutually exclusive events A and B is $P(A \cup B) = P(A) + P(B)$.

c. Given that events A and B are independent, and that $P(A) = 0.9$ and $P(B|A) = 0.5$, then $P(A \cap B) = 0.45$.

d. A bivariate distribution is a distribution is a joint probability distribution of two variables.

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Events A and B are independent

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

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Events A and B are independent

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

FALSE: since $0.3 * 0.5 = 0.15 \neq 0.8$

Question 1(b)

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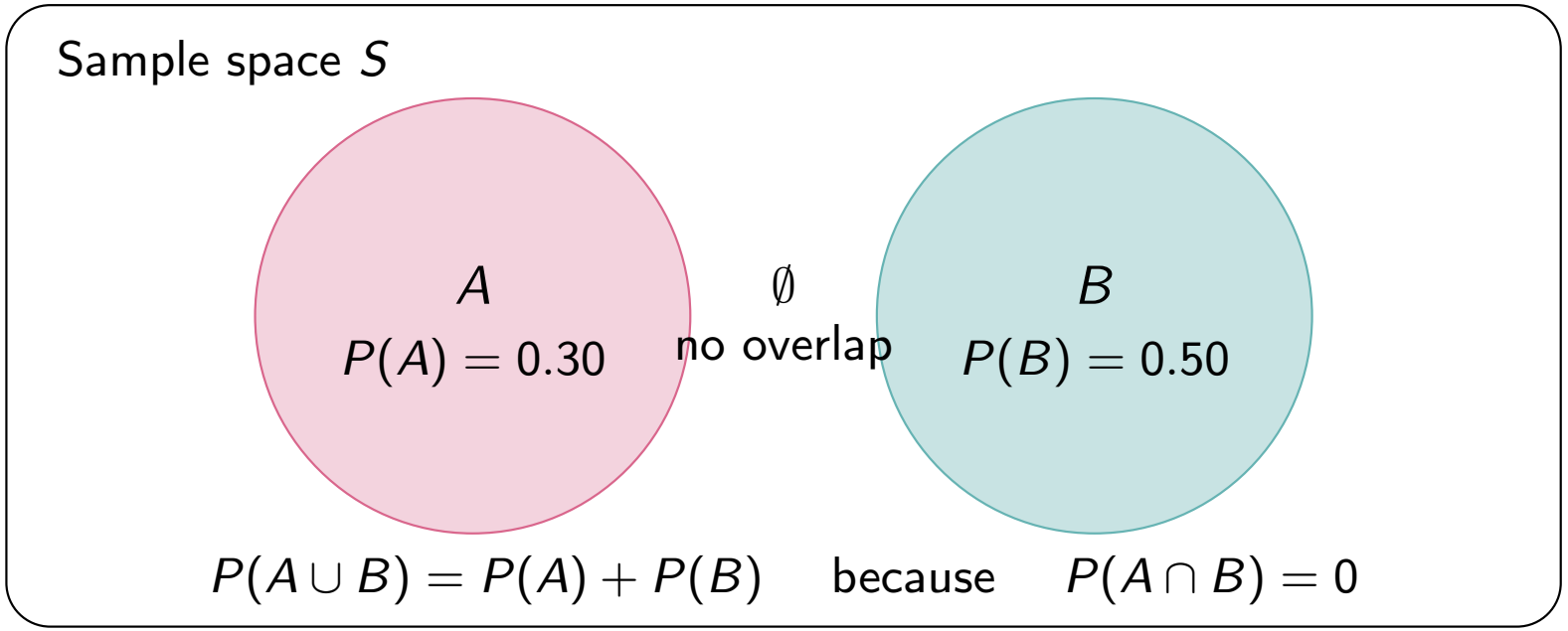

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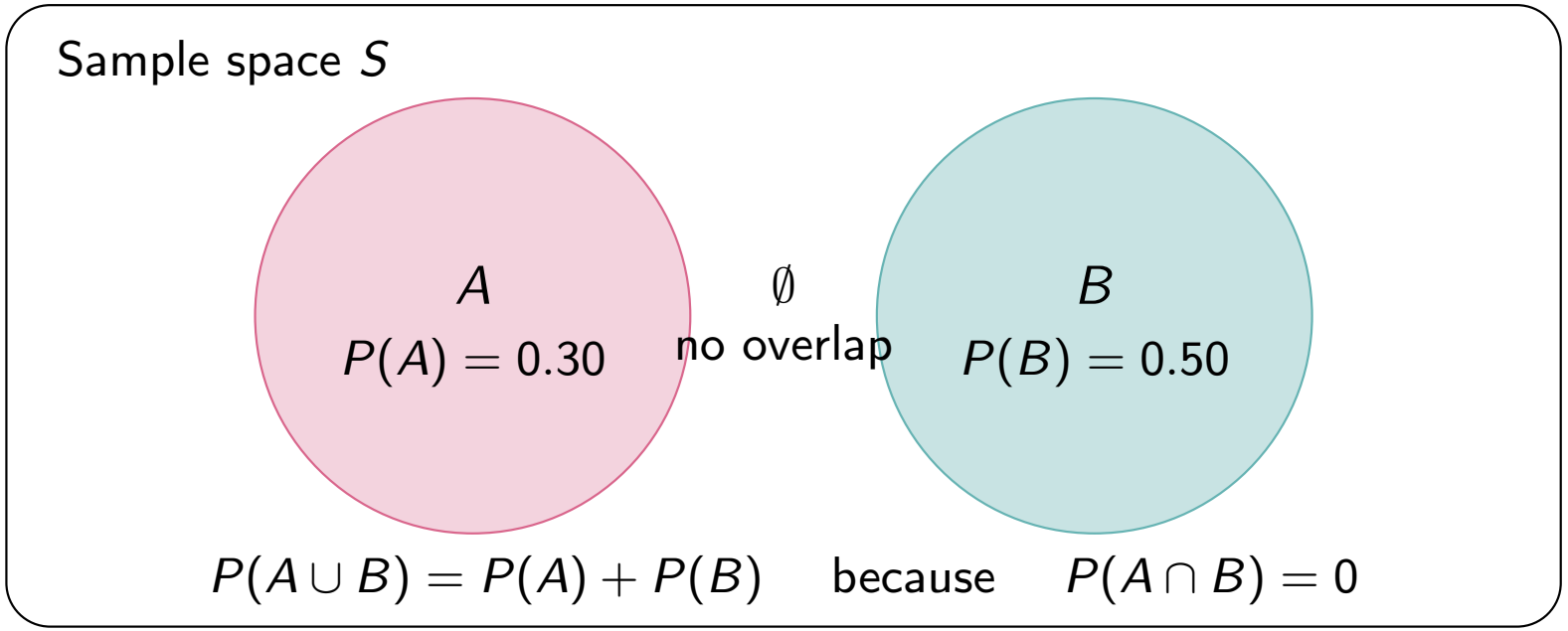

TRUE

Question 1(c)

Statement: *Given that events A and B are independent, and that $P(A) = 0.9$ and $P(B|A) = 0.5$, then $P(A \cap B) = 0.45$.*

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TRUE: since $P(A \cap B) = P(B|A) \cdot P(A) = P(B) \cdot P(A) = 0.9 * 0.5 = 0.45$

Question 1(d)

Statement: *A bivariate distribution is a distribution is a joint probability distribution of two variables.*

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TRUE: by definition.

Question 2

In a country with one million voters, the Conservative and Progressive political parties contested elections in 2017 and 2020. In 2017, the Progressive party won with 55% of the vote. 350,000 people who voted Progressive in 2017 also voted Progressive in 2020. 100,000 people who voted Conservative in 2017 also voted Conservative in 2020.

a. Who won the election in 2020, and how many votes did they receive?
b. What is the probability that a randomly selected person voted Progressive in both elections?
c. What is the probability that a randomly selected person voted Progressive in 2017 but did not vote Progressive in 2020?
d. Given someone who voted Progressive in 2017, what is the probability that this person voted for the same party in 2020?
e. Given someone who voted Conservative in 2017, what is the probability that this person voted for the other party in 2020?

Question 2

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- Information in a 2×2 Contingency Table

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020			
Conservative 2020			
Total			

- $\rightarrow$ Bivariate distribution

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	350,000	350,000	700,000
Conservative 2020	200,000	100,000	300,000
Total	550,000	450,000	1,000,000

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(a) Who won the election in 2020, and how many votes did they receive?

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	350,000	350,000	700,000
Conservative 2020	200,000	100,000	300,000
Total	550,000	450,000	1,000,000

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(a) Who won the election in 2020, and how many votes did they receive?

`The Progressive party won the 2020 election with 700,000 votes`

(b) What is the probability that a randomly selected person voted Progressive in both elections?

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	350,000	350,000	700,000
Conservative 2020	200,000	100,000	300,000
Total	550,000	450,000	1,000,000

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(a) Who won the election in 2020, and how many votes did they receive?

`The Progressive party won the 2020 election with 700,000 votes`

(b) What is the probability that a randomly selected person voted Progressive in both elections?

$Pr(P \ \ in \ \ 2020 \cap P \ \ in \ \ 2017) = 0.35$

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
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(c) What is the probability that a randomly selected person voted Progressive in 2017 but did not vote Progressive in 2020?

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	Progressive 2017	**Conservative 2017**	**Total**
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Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(c) What is the probability that a randomly selected person voted Progressive in 2017 but did not vote Progressive in 2020?

$Pr(C \ \ in \ \ 2020 \cap P \ \ in \ \ 2017) = 0.2$

(d) Given someone who voted Progressive in 2017, what is the probability that this person voted for the same party in 2020?

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(c) What is the probability that a randomly selected person voted Progressive in 2017 but did not vote Progressive in 2020?

$Pr(C \ \ in \ \ 2020 \cap P \ \ in \ \ 2017) = 0.2$

(d) Given someone who voted Progressive in 2017, what is the probability that this person voted for the same party in 2020?

$Pr(P20|P17) = \frac{Pr(P20 \cap P17)}{Pr(P17)} = \frac{0.35}{0.55} = 0.636$

(e) Given someone who voted Conservative in 2017, what is the probability that this person voted for the other party in 2020?

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(c) What is the probability that a randomly selected person voted Progressive in 2017 but did not vote Progressive in 2020?

$Pr(C \ \ in \ \ 2020 \cap P \ \ in \ \ 2017) = 0.2$

(d) Given someone who voted Progressive in 2017, what is the probability that this person voted for the same party in 2020?

$Pr(P20|P17) = \frac{Pr(P20 \cap P17)}{Pr(P17)} = \frac{0.35}{0.55} = 0.636$

(e) Given someone who voted Conservative in 2017, what is the probability that this person voted for the other party in 2020?

$Pr(P20|C17) = \frac{Pr(P20 \cap C17)}{Pr(C17)} = \frac{0.35}{0.45} = 0.777$

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(f) Do the probabilities calculated in parts (d) and (e) sum to 1? Should they? Explain why or why not.

Question 2

	Progressive 2017	**Conservative 2017**	**Total**
Progressive 2020	0.35	0.35	0.70
Conservative 2020	0.20	0.10	0.30
Total	0.55	0.44	1

(f) Do the probabilities calculated in parts (d) and (e) sum to 1? Should they? Explain why or why not.

No, nor must they. We are dealing with probability statements about separate sub-populations here. In (d) the sub-population who voted Progressive in 2017 and in (e) the sub-population who voted Conservative in 2017

(g) Draw a probability tree to compute all the possible joint probabilities.

Question 3

At the Queen Victoria Market, a poultry producer is selling ducklings from a cage that contains 3 males and 5 females. You ask her to select two ducklings for you at random. Let A be the event that the first duckling selected is male and let B be the event that both ducklings have the same gender.

a. Calculate $P(A)$, $P(B)$ and $P(B|A)$.

Question 3

At the Queen Victoria Market, a poultry producer is selling ducklings from a cage that contains 3 males and 5 females. You ask her to select two ducklings for you at random. Let A be the event that the first duckling selected is male and let B be the event that both ducklings have the same gender.

a. Calculate $P(A)$, $P(B)$ and $P(B|A)$.

$$\Pr(A) = \frac{3}{8}$$

$$\Pr(B) = \frac{3}{8} \times \frac{2}{7} + \frac{5}{8} \times \frac{4}{7} = \frac{26}{56} = 0.464$$

For $\Pr(B)$, the two probabilities added together are the probability that both are male, plus the probability that both are female.

$$\Pr(B \mid A) = \frac{\Pr(A \cap B)}{\Pr(A)} = \frac{3/8 \times 2/7}{3/8} = \frac{2}{7} = 0.286$$

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b. Are events A and B independent? Explain why or why not.

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   Not independent

b. Are events A and B mutually exclusive? Explain why or why not.

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  Not independent

- b Are events A and B mutually exclusive? Explain why or why not.

  The events are not mutually exclusive as both events can occur at the same time if both ducklings are male.

Question 4

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Begin by defining the events:

- W: The employee selected is a woman.
- M: The employee selected is married.

We are given $P(W) = 0.3$ and $P(M \cap W) = 0.06$. Thus,

$$P(M \mid W) = \frac{P(M \cap W)}{P(W)} = \frac{0.06}{0.03} = 0.2$$

Question 5

The joint distribution of two random variables X and Y is given below. Tabulated values are of the form $P(X = x \text{ and } Y = y)$.

$y \backslash x$	0	1
0	0	1/3
1	1/4	5/12

Are X and Y independent?

Question 5

The joint distribution of two random variables X and Y is given below. Tabulated values are of the form $P(X = x \text{ and } Y = y)$.

$y \backslash x$	0	1
0	0	1/3
1	1/4	5/12

Are X and Y independent?

$$\mathrm{COV}(X, Y) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i y_j \; p(x_i, y_i) - \mu_x \mu_y$$

$$\mu_x = \sum_{i=1}^{k} x_i \; p(x_i)$$

Question 5

$y \backslash x$	0	1
0	0	1/3
1	1/4	5/12

$$\mathrm{COV}(X, Y) = \sum_{x=0}^{1} \sum_{y=0}^{1} xy \; P(X = x_i \cap Y = y) - \mu_X \mu_Y$$

Thanks for your attention!

Let me know if there is any questions!